

Microcontroller Lab 2: C program in CCS

- Objectives:**
- Further Understanding of IDE. **Code Composer Studio – the debug environment , single stepping**
 - Blinking an inbuilt LED on P1.0 continuously, 5sec ON, 5sec OFF time to be correct to 10%
 - Turning the inbuilt LED on P1.6 ON at the push of the inbuilt push-button, LED turns OFF when button is released
 - Turning the inbuilt LED on P1.6 ON at the push of the inbuilt push-button, LED remains ON when button is released
 - Toggling the state of the inbuilt LED on P1.6 at each push of the inbuilt push-button

Things to do:

1. Refer to the manual/literature attached with reference to the LED blink task. **You can also open the example file of the LED blink in CCS.** Familiarize yourself with the perspectives in CCS, and the different drop-down menu options. How do you create a new project, write code (not needed if example project is loaded), compile and load the program with or without debug options.
2. Go through the lines of code and understand what each line means, to answer questions that follow.
3. Your lab records must have answers to the questions, along with submission online of screenshots of the IDE at different stages.
4. In the lab, you will also have to demonstrate the outputs on hardware for evaluation.

Questions to be answered for lab 1. **You will not be allowed to conduct the experiment if you do not answer these questions**

1. To which pins on the micro-controller is the inbuilt LEDs and push button connected?
2. What is the principle used to generate the time delays required? What are the advantages and disadvantages? (see the information on the compile window for this. What does this mean?)
3. How is the duration of delay computed, given the clock frequency?
4. What needs to be added in code to cause LED to turn OFF on button release?
5. How is the setting of the LED permanently at the first push of the button done?
6. How is toggling of the LED achieved? Why is a delay required in the toggle program?

Outputs to be shown.

1. Blinking an inbuilt LED continuously
2. Turning the inbuilt LED ON at the push of the inbuilt push-button, LED remains ON when button is released
3. Turning the inbuilt LED ON at the push of the inbuilt push-button, LED turns OFF when button is released
4. Toggling the state of the inbuilt LED at each push of the inbuilt push-button

Resources required:

Microcontroller board, laptops for code development